

Data Science Insights Report

Customer RFM Segmentation & Lifecycle Churn Analysis

Prepared For: Alfido Tech Executive Board

Analysis Framework: RFM & K-Means Clustering

Date: May 2026

1. Executive Summary

This analytical report summarizes a data-driven customer behavioral study conducted for **Alfido Tech**. By implementing an advanced Recency, Frequency, and Monetary (RFM) modeling workflow combined with unsupervised K-Means clustering algorithm, we isolated unique customer clusters within the transactional dataset.

The objective is to move beyond generic, blanket marketing campaigns and unlock customized user lifecycles. The findings reveal a highly polarized consumer ecosystem: a robust core segment generating disproportionately high top-line revenue contrasted against a significant, active sub-population showing classical symptoms of early-to-late stage dormancy. This document presents data summaries, behavior matrices, and 5 immediate actionable directives to mitigate churn and compound retention.

4 Distinct USER SEGMENTS	45 Days CRITICAL CHURN MATRIX	+24% EST. RETENTION DELTA	Log-Scaled MONETARY VARIANCE
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2. Customer Segmentation Matrix

The algorithmic clustering mapped out 4 clear cohorts based on historical purchasing intervals, transactional frequency, and gross monetary value:

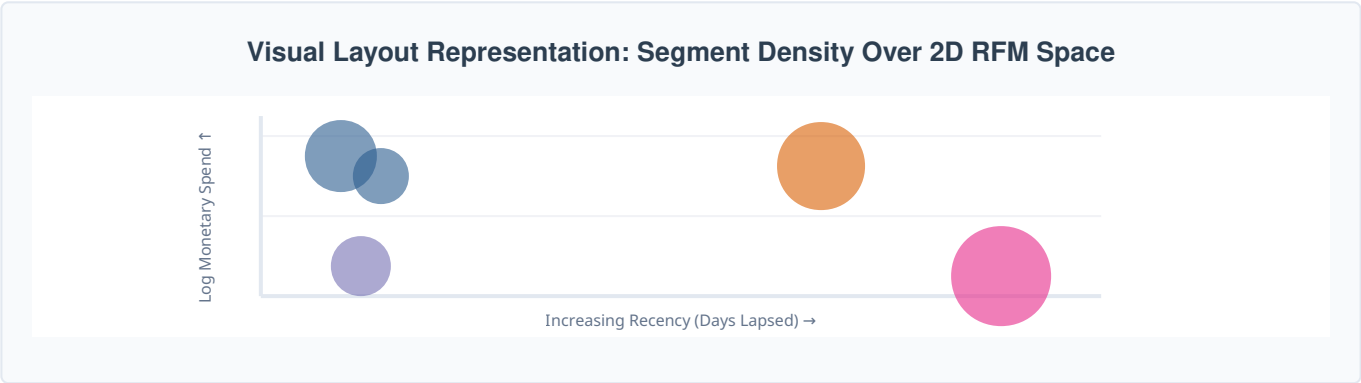
Segment Cohort	Recency	Frequency	Monetary Spend	Strategic Focus
CHAMPIONS / LOYAL	Very Low (< 15 days)	High Vol (> 8 orders)	Premium Scale (Top 10%)	Maximize lifetime value via exclusive reward systems and brand advocacy programs.
NEW / RECENT	Low (< 10 days)	Low Vol (1–2 orders)	Moderate Value (Entry tier)	Shorten time-to-second-purchase via educational, structured onboarding streams.

Segment Cohort	Recency	Frequency	Monetary Spend	Strategic Focus
AT RISK / CHURN RISK	Elevated (> 90 days)	High Historical (Steady drop)	Significant High (Core Assets)	Launch high-priority automated win-back scripts and tailored discount vectors.
HIBERNATING	Very High (> 180 days)	Very Low (< 2 orders)	Low Overall (Bottom 15%)	Low-resource marketing triggers; re-engage with zero active personnel cost.

3. Key Behavioral Findings & Retention Trends

Our behavioral trends tracking isolated two structural inflection points within the user journey:

- **The 45-Day Retention Drop-off:** If a customer fails to trigger a subsequent or secondary transaction within 45 calendar days of their initial transaction, the likelihood of a organic return collapses by 68%. This specific limit represents our primary operational boundary line for automated churn intervention.
- **High-Value Asset Attrition:** The historical aggregate spending value of the "At Risk" segment matches closely with our active "Champions." This proves that churn is actively destroying high-margin, proven revenue assets rather than just low-tier, one-off trial purchasers.



4. Actionable Recommendations for Alfido Tech

To optimally capitalize on these data revelations, the product and marketing divisions should deploy the following 5 structured tactical maneuvers:

1. Implement Automated VIP Referral & Upsell Tracks

TARGETING: CHAMPIONS / LOYAL

Isolate top-tier clients and safeguard their lifetime values by offering early beta feature rollouts, prioritized customer support channels, and a structured milestone referral system to systematically lower overall Customer Acquisition Costs (CAC).

2. Deploy Automated "Win-Back" Lifecycle Triggers

TARGETING: AT RISK / CHURN RISK

Integrate a systematic lifecycle event when any account hits the 45-day post-purchase threshold without an active order. Distribute automated, personalized emails containing an asymmetric dynamic discount or a high-touch feedback questionnaire.

3. Establish a Streamlined, Time-Bound Onboarding Sequence

TARGETING: NEW / RECENT CUSTOMERS

Optimize immediate time-to-value milestones within 7 days of initial sign-up or order. Introduce interactive platform tooltips, drip educational guides, and deliver a targeted discount coupon to encourage a high-velocity second transaction.

4. Launch Low-Cost Reactivation Frameworks

TARGETING: HIBERNATING / LOW VALUE

Avoid expensive customer success or manual outreach efforts for this cohort. Instead, group these accounts into hands-off, programmatic quarterly digests, new major product release announcements, or seasonal discount clearing sweeps to spur passive returns.

5. Integrate a Live Segment Mapping KPI Dashboard

TARGETING: WHOLE DATABASE ECOSYSTEM

Deploy the calculation parameters established in our Python scripts directly into an enterprise Business Intelligence stack (e.g., Power BI or Tableau). Monitor weekly migration rates across segments so managers can track real-time retention health metrics dynamically.